

B2B Hub Modernization

Portal Routing Requirements

PMO-5000 · Phase 1 · Routing Configuration Initiative

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Document Type	Business Requirements Document (BRD)
Epic	PMO-5000 — B2B Hub Modernization & CMS Initiative
Sub-Initiative	Portal Routing Configuration — Segment × Contract × State Mapping
Phase	Phase 1 (April – June 2026)
Status	Draft — Pending Stakeholder Review
Approvers	Ariel Goldman (B2B Business Lead), Manny Steigman (Senior Stakeholder)
Primary Owner	Mr. Smith — Senior Product Manager, B2B Hub
Engineering POC	Shia Bernstein (B2B Technical Lead), Brendan Bagley (Backend PM)
Change Process	DES (Data Entry System) or Jira ticket submitted to WebOps

1. Purpose & Scope

This document defines requirements for the Portal Routing Configuration initiative within the B2B Hub Modernization program (PMO-5000). It specifically addresses how the system determines which BDBTools portal a B2B customer is directed to — based on the intersection of their **organization segment**, the **cooperative contract** they are accessing, and their **state** of operation.

Previously, the CMS prototype included a PortalMappingsEditor UI allowing administrators to configure these mappings via the admin panel. Based on stakeholder alignment, this capability is **removed from the CMS scope**. Portal routing configuration will instead be managed through a controlled operational process: either via the {B('Data Entry System (DES)')} or through a **Jira ticket submitted to the WebOps team**.

- Portal routing rules directly affect which purchasing portal a customer lands on. Misconfiguration can result in customers accessing incorrect pricing tiers, ineligible contract portals, or complete routing failure. Changes must follow the defined change management process.

1.1 In-Scope

- Definition of the Segment × Contract × State routing data model
- Lookup priority logic (resolution order for matching rules)
- Change management process: DES entries and WebOps Jira ticket workflow
- Audit and traceability requirements for routing changes
- Validation rules and error handling for routing configuration
- Operational runbook requirements for WebOps team
- Integration contract between routing data store and the B2B Hub front-end
- Reporting/visibility requirements for current routing state

1.2 Out of Scope

- **CMS admin UI for portal mapping** — removed from Phase 1; deferred to Phase 2 or permanently delegated to DES/WebOps
- BDBTools portal parameter management (read-only in Phase 1; owned by Sam Herman)
- New portal creation — adding a new portal ID to BDBTools is a BDBTools workflow
- Authenticated/logged-in portal experience (Phase 2)
- Automated domain-based routing (PMO-5000 domain sniffing — separate initiative)

2. Background & Current State

2.1 The Routing Problem

The B2B Hub serves 12+ organization segments across cooperative contracts and custom portals. Each contract may serve multiple segments, and within a segment, the correct portal can vary by state — for example, a State/Local Government customer in California accessing the Omnia Partners contract may be directed to a different portal than the same segment in Texas.

Today, all state-to-contract-to-portal routing is hard-coded in the application. Any change — adding a new state association, updating a portal ID, or adjusting segment eligibility — requires a developer ticket, QA pass, and production deployment. Estimated lead time is 2–8 weeks for even minor routing updates.

2.2 Why CMS Control Was Removed

An earlier CMS prototype iteration included a PortalMappingsEditor component allowing Ariel Goldman and authorized admins to directly configure segment × contract × state mappings through the admin panel. This approach was reviewed and rejected by stakeholders for the following reasons:

- **Risk profile:** Portal routing errors are high-impact — they affect revenue, contract compliance, and customer access. Direct CMS editing introduces operator error risk without a downstream validation gate.
- **Change velocity:** Routing changes are low-frequency but high-consequence. They do not benefit from the self-service model that drives CMS value for content updates.
- **Audit requirements:** Contract and government-segment portals have traceability obligations that are better served by a ticketing workflow than a CMS edit history.
- **Separation of concerns:** Content editors (Ariel Goldman, Josh Bernstein) should not have access pathways to routing infrastructure that can misdirect all customers on a given contract.

3. Routing Data Model

3.1 Core Entity: Portal Mapping Rule

A portal mapping rule defines the BDBTools portal ID to use when a customer matches a specific combination of contract, segment, and state. Each rule is a row in the routing configuration table.

Field	Type	Description
rule_id	UUID	System-generated primary key. Immutable after creation.
contract_id	VARCHAR(64)	References a cooperative contract (e.g. omnia, ei, tips, buyboard, equalis, mhcc).
segment	VARCHAR(64)	B2B organization segment key. Empty string ("") = any segment (wildcard).
state	CHAR(2)	Two-letter US state code. Empty string ("") = national / all states (wildcard).
portal_id	VARCHAR(16)	BDBTools portal ID integer (e.g. "944"). Must be a valid, active portal in BDBTools.
label	VARCHAR(128)	Human-readable label for ops/audit purposes (e.g. "Omnia CA State Gov").
active	BOOLEAN	Whether this rule is currently active. Inactive rules are ignored during lookup.

Field	Type	Description
effective_from	TIMESTAMP	UTC timestamp when this rule becomes active. Supports scheduled routing changes.
effective_until	TIMESTAMP	UTC timestamp when this rule expires. NULL = no expiry.
created_by	VARCHAR(64)	Identity of operator who created the rule (DES user ID or Jira ticket reference).
created_at	TIMESTAMP	UTC creation timestamp.
updated_by	VARCHAR(64)	Identity of last operator to modify the rule.
updated_at	TIMESTAMP	UTC last-modified timestamp.
jira_ticket	VARCHAR(32)	WebOps Jira ticket ID that authorized this change (e.g. WEBOPS-1042).
notes	TEXT	Optional free-text notes for operational context.

3.2 Segments Reference

The following segment keys are recognized by the routing system. All 12 segments must be supported. The empty string wildcard applies to any segment not explicitly matched.

Segment Key	Display Name	Description
federal	Federal	U.S. Federal Government agencies
higher-ed	Higher Education	Colleges, universities, community colleges
state-local	State/Local Government	State agencies, municipalities, counties
k12	K-12 Education	Public and private K-12 schools and districts
diversity	Diversity Suppliers	Certified diverse business enterprises
smb	Small-Medium Business	Non-government commercial small/mid businesses
corporate	Corporate	Large commercial/enterprise accounts
faith	Faith Organizations	Religious and faith-based organizations
creative	Creative/Production & Post	Media production studios and post houses
systems-integrator	Systems Integrator	IT integrators and resellers
healthcare	Healthcare	Hospitals, clinics, healthcare systems
nonprofit	Non-Profit	501(c)(3) and similar nonprofit organizations
"" (empty)	(Wildcard)	Matches any segment — used as a fallback rule

3.3 Contracts Reference

The following cooperative contracts are currently active in the B2B Hub. Each contract maps to one or more BDBTools portals depending on segment and state.

Contract Key	Display Name	Notes
omnia	Omnia Partners	Cooperative purchasing for public sector
ei	E&I Cooperative	Higher education cooperative
tips	TIPS-USA	National cooperative purchasing
buyboard	BuyBoard	Texas-based cooperative
equalis	Equalis Group	Public sector cooperative
mhec	MHEC	Higher education consortium — regional

4. Lookup Priority Logic

When the B2B Hub needs to resolve a portal for a given customer, it evaluates the routing table using a defined priority cascade. The first matching rule wins. This logic is identical to the existing `resolvePortalMapping()` function in the front-end prototype and must be preserved in the back-end routing service.

Priority	Match Type	Example Rule Values	When to Use
1 (Highest)	Exact State + Exact Segment	state = 'CA', segment = 'k12'	Most specific rule. Use when a contract has state-specific portal variants for a particular segment.
2	Exact State + Any Segment	state = 'CA', segment = ''	State-level override regardless of segment. Use when all segments in a state share one portal.
3	All States + Exact Segment	state = '', segment = 'k12'	National default for a specific segment. Use when a segment always uses one portal nationwide.
4 (Lowest)	All States + Any Segment (Global Fallback)	state = '', segment = ''	Catch-all for the contract. Every contract must have exactly one global fallback rule.
— (Error)	No Match	N/A	If no rule matches and no global fallback exists, the system must surface a configuration error. Do not silently fail.

- ✓ Every active contract must have at least one global fallback rule (state = "", segment = ""). This ensures no customer reaches a routing error for an active contract. The DES and WebOps intake process must validate this constraint before applying any change.

4.1 Lookup Algorithm (Pseudocode)

```

function resolvePortal(contractId, state, segment):
  rules = getActiveRules(contractId) // filter by active=true, within effective dates

  // Priority 1: Exact state + exact segment
  match = rules.find(r => r.state == state AND r.segment == segment
  AND r.state != '' AND r.segment != '')
  if match: return match.portal_id

  // Priority 2: Exact state + any segment
  match = rules.find(r => r.state == state AND r.segment == ''
  AND r.state != '')
  if match: return match.portal_id

  // Priority 3: Any state + exact segment
  match = rules.find(r => r.state == '' AND r.segment == segment
  AND r.segment != '')
  if match: return match.portal_id

  // Priority 4: Global fallback
  match = rules.find(r => r.state == '' AND r.segment == '')
  if match: return match.portal_id

  // Error: no rule found
  throw RoutingConfigurationError('No portal mapping for: ' + contractId)

```

5. Change Management Process

Portal routing configuration changes are operational changes — not content changes. They must follow a controlled workflow to ensure traceability, validation, and rollback capability. Two pathways are supported:

5.1 Pathway A — Data Entry System (DES)

The DES is the preferred pathway for routine portal routing updates performed by authorized B2B operations personnel. DES users must have a designated role that grants access to the routing configuration module.

#	Actor	Action
1	Requestor	Identifies routing change need (new state, portal ID update, segment eligibility change)
2	Requestor	Logs into DES and navigates to Portal Routing Configuration module
3	DES	Displays current active rules for the selected contract
4	Requestor	Creates new rule or modifies/deactivates existing rule
5	DES	Validates: portal ID exists in BDBTools, no duplicate rules at same priority, global fallback present
6	DES	Requires entry of change justification and associates with a Jira ticket ID (optional but recommended)

#	Actor	Action
7	DES Approver	Second authorized user reviews and approves the change (four-eyes principle for contract/gov segments)
8	DES	Applies change to routing table; records created_by, created_at, jira_ticket in audit log
9	DES	Triggers cache invalidation for affected contract routing config
10	QA	Requestor or designee verifies routing in staging/production using the routing test utility

5.2 Pathway B — WebOps Jira Ticket

The Jira pathway is used when the requestor does not have DES access, when the change requires WebOps engineering involvement, or when the change is part of a larger deployment. This pathway also applies during any period when DES routing module is unavailable.

#	Actor	Action
1	Requestor (B2B PM or Ops)	Opens Jira ticket in the WebOps project. Uses the Portal Routing Change request type.
2	Requestor	Completes all required fields in the ticket template (see Section 5.3).
3	B2B Business Lead (Ariel Goldman)	Reviews and approves the ticket. Adds approval comment.
4	Manny Steigman	Required approver for any change affecting customer-facing government or cooperative contract portals.
5	WebOps Engineer	Implements the routing change via DES or direct data store update.
6	WebOps Engineer	Records the jira_ticket field in the routing rule audit log.
7	WebOps Engineer	Marks Jira ticket Done; links to specific rule_id(s) modified.
8	Requestor	Validates routing behavior in production. Closes ticket or raises follow-up if incorrect.

5.3 Jira Ticket Template — Required Fields

All WebOps portal routing tickets must include the following fields. Incomplete tickets will be returned to the requestor before any action is taken.

Field	Description	Requirement
Change Type	ADD new rule / UPDATE existing rule / DEACTIVATE rule	Required
Contract ID	Cooperative contract key (e.g. omnia, ei, tips)	Required
Segment	Segment key or "All Segments" (wildcard)	Required
State	Two-letter state code or "All States" (wildcard)	Required

Field	Description	Requirement
Portal ID (New)	BDBTools portal ID to route to	Required for ADD/UPDATE
Portal ID (Previous)	Current portal ID being replaced (for traceability)	Required for UPDATE
Rule Label	Human-readable label for the new rule	Required for ADD
Effective Date	Date/time change should take effect (UTC). Default: immediately.	Optional
Business Justification	Why this change is needed. Include contract documentation if applicable.	Required
Requestor	Name and email of person requesting the change	Required
Approved By	Name of B2B Business Lead approver (Ariel Goldman or delegate)	Required
Manny Steigman Approval	Confirmation of approval for government/cooperative segments	Required for gov/coop
Staging Verified	Yes/No — has this been tested in staging?	Required
Rollback Plan	What to do if the change causes routing failures in production	Required

6. Validation Rules

The following validation rules must be enforced by DES and by any back-end service that writes to the routing configuration table. Violations must block the change and return a descriptive error.

Rule ID	Name	Severity	Description
V-01	Portal ID Existence	BLOCK	portal_id must resolve to an active portal in BDBTools. If the portal does not exist or is inactive, the rule must be rejected.
V-02	Duplicate Rule	BLOCK	No two active rules for the same contract_id may share the same (state, segment) pair. Deactivate the existing rule first.
V-03	Global Fallback Present	BLOCK	Every active contract must retain at least one global fallback rule (state="", segment="). A change that would remove the last fallback must be blocked.
V-04	Contract Key Validity	BLOCK	contract_id must be one of the recognized contract keys. Unknown keys are rejected.
V-05	Segment Key Validity	BLOCK	segment must be one of the 12 recognized segment keys or empty string. Unknown segment keys are rejected.
V-06	State Code Validity	BLOCK	state must be a valid two-letter US state code or empty string. Invalid codes are rejected.

Rule ID	Name	Severity	Description
V-07	Effective Date Logic	BLOCK	If effective_until is set, it must be after effective_from. Zero or negative duration windows are rejected.
V-08	Four-Eyes Approval	WARNING	For contracts serving Federal, Higher Ed, State/Local, or K-12 segments, a second-approver sign-off is required in DES. Warning only in DES; ticket pathway enforces via workflow.
V-09	Jira Ticket Reference	WARNING	jira_ticket field is strongly recommended for all changes. System emits a warning if left empty but does not block.
V-10	Cascade Orphan Check	WARNING	If deactivating a rule that was the only state-specific rule for a contract, warn that customers in that state will now fall through to the national fallback.

7. Audit & Traceability Requirements

Given that portal routing affects contract-level and government-segment access, full audit traceability is required for all routing configuration changes.

AUD-01: Immutable Audit Log

Every INSERT, UPDATE, or deactivation of a routing rule must be written to an immutable audit log. Log entries cannot be deleted or modified.

AUD-02: Audit Log Fields

Each log entry must capture: rule_id, contract_id, segment, state, portal_id (before), portal_id (after), change_type, operator_id, timestamp (UTC), jira_ticket, change_justification.

AUD-03: Retention

Audit log entries must be retained for a minimum of 3 years.

AUD-04: Read Access

B2B Product Management, B2B Business Lead, and WebOps leads must have read access to the full audit log. No routing change should be invisible to these roles.

AUD-05: Diff View

The DES routing module and WebOps tooling must display a before/after diff for any pending change before it is committed.

AUD-06: Routing Snapshot

The system must support exporting the current routing table state as a timestamped snapshot (CSV or JSON) for compliance and change-management documentation.

8. Integration Requirements

8.1 Front-End / Hub Integration

The B2B Hub must resolve portal routing at sign-up time (Sign-Up Tray flow) by calling the routing service with the customer's identified contract, segment, and state. The following requirements apply:

- The front-end `resolvePortalMapping()` function must be replaced by an API call to the routing service — routing rules must not be embedded in front-end code.
- The routing API must return the resolved `portal_id` and label within 200ms at p95 to avoid sign-up flow latency.
- The front-end must handle a `RoutingConfigurationError` gracefully — display a fallback message and surface a contact path to B2B support. Do not white-screen.
- Routing results may be cached per session. They must not be cached across sessions or users.
- The routing API must be versioned (e.g. `/v1/routing/resolve`) to allow non-breaking evolution.

8.2 BDBTools Integration

BDBTools remains read-only in Phase 1. The routing service must validate portal IDs against BDBTools at rule-creation time. At runtime, the routing service only returns a `portal_id` — BDBTools is invoked separately by the existing portal parameter resolution layer.

8.3 Routing Test Utility

A routing test utility must be provided — accessible to B2B Product Management and WebOps — that accepts `contract_id`, `segment`, and `state` as inputs and returns the resolved `portal_id`, the matching rule, and the priority level used. This tool is essential for post-change validation.

9. Roles & Permissions

Role	Current Assignee(s)	Routing Permissions
B2B Business Lead	Ariel Goldman	Can VIEW routing table. Can REQUEST changes. Can APPROVE Jira tickets. Cannot directly write routing rules.
Senior Stakeholder	Manny Steigman	Required approver for government and cooperative contract routing changes. Approval required before any customer-facing routing deployment.
B2B Product Manager	Mr. Smith	Can VIEW routing table and audit log. Can REQUEST and draft changes. Owns requirements and process. Cannot directly write routing rules.
DES Routing Operator	Designated ops personnel	Can CREATE, UPDATE, DEACTIVATE routing rules via DES. Subject to four-eyes validation for high-impact segments.
DES Routing Approver	Designated approver	Second-approver role in DES for government/cooperative segment changes.
WebOps Engineer	WebOps team	Can implement routing changes via DES or direct data store access when executing an approved Jira ticket.

Role	Current Assignee(s)	Routing Permissions
BDBTools Owner	Sam Herman	Owns BDBTools portal parameter data. Consulted when a new portal ID is needed. Not involved in routing rule management.
Read-Only / Viewer	Josh Bernstein, B2B team	Can VIEW current routing table and audit log. No write access.

10. Example Routing Rules

The following illustrates a sample routing table for the Omnia Partners contract across segments and states. Portal IDs are illustrative placeholders — actual values must be sourced from BDBTools.

Contract	Segment	State	Portal ID	Label	Note
omnia	state-local	CA	901	Omnia CA State/Local	Priority 1 — exact state+segment
omnia	state-local	TX	902	Omnia TX State/Local	Priority 1 — exact state+segment
omnia	k12	CA	903	Omnia CA K-12	Priority 1 — exact state+segment
omnia	state-local		910	Omnia National State/Local	Priority 3 — national segment default
omnia	k12		911	Omnia National K-12	Priority 3 — national segment default
omnia	higher-ed		912	Omnia National Higher Ed	Priority 3 — national segment default
omnia			900	Omnia Global Fallback	Priority 4 — global fallback (required)

11. Open Questions & Decisions Needed

ID	Topic	Question	Owner	Status
OQ-01	DES Routing Module	Does DES currently have a routing configuration module, or does one need to be built? Who owns DES development?	Brendan Bagley / WebOps	Open
OQ-02	Four-Eyes Approval	Who is the designated DES Routing Approver role for government segment changes when Manny Steigman is unavailable?	Ariel Goldman / Manny Steigman	Open
OQ-03	Ephraim Pushback	Ephraim is expected to push back on scope of routing changes. What is the minimum viable routing service for Phase 1?	Mr. Smith / Shia Bernstein	Pending

ID	Topic	Question	Owner	Status
OQ-04	Existing Customer Routing	Current lean: apply new routing logic to new sign-ups only. Existing customers remain on hard-coded routing. Needs formal decision.	Mr. Smith / Ariel Goldman	Open
OQ-05	Staging Environment	Is there a staging environment where routing changes can be validated before production? Requirement for Staging Verified field in tickets.	Shia Bernstein	Open
OQ-06	Cache Invalidation	What is the cache TTL for routing configuration? Who triggers cache invalidation when DES commits a change?	Shia Bernstein / WebOps	Open

12. Approvals

This document requires sign-off from the following stakeholders before implementation begins.

Name	Role	Team	Signature	Date
Ariel Goldman	B2B Business Lead	B2B Product	_____	_____
Manny Steigman	Senior Stakeholder	B2B Product	_____	_____
Shia Bernstein	B2B Technical Lead	Engineering	_____	_____
Brendan Bagley	Backend PM	Engineering	_____	_____
Mr. Smith	Senior PM — B2B Hub	Product	_____	_____